

$$\vec{r} = r \hat{r}$$

$$\dot{\vec{r}} = \dot{r} \hat{r} + r \dot{\theta} \hat{\theta}$$

$$T = \frac{1}{2} m \|\dot{\vec{r}}\|^2$$

$$= \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2)$$

$$\frac{\partial T}{\partial r} = m r \dot{\theta}^2 \quad \frac{\partial T}{\partial \theta} = 0$$

$$\frac{\partial T}{\partial \dot{r}} = m \dot{r} \quad \frac{\partial T}{\partial \dot{\theta}} = m r^2 \dot{\theta}$$

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{r}} = m \ddot{r} \quad \frac{d}{dt} \frac{\partial T}{\partial \dot{\theta}} = m r^2 \ddot{\theta} + 2 m r \dot{r} \dot{\theta}$$

$$r = R, \quad \dot{r} = 0, \quad \ddot{r} = 0 \Rightarrow$$

$$\frac{\partial T}{\partial r} = m R \dot{\theta}^2, \quad \frac{d}{dt} \frac{\partial T}{\partial \dot{r}} = 0$$

$$\frac{\partial T}{\partial \theta} = 0, \quad \frac{d}{dt} \frac{\partial T}{\partial \dot{\theta}} = m R^2 \ddot{\theta}$$

$$Q = \vec{F} \cdot \frac{\partial \vec{r}}{\partial \vec{q}}$$

$$\vec{F}_g = -mg \cos \theta \hat{n} - mg \sin \theta \hat{\theta}$$

$$\vec{F}_N = -F_N \hat{n}$$

$$\vec{F}_p = F_p \hat{\theta}$$

$$\vec{F}_D = -cR\dot{\theta} \hat{\theta}$$

$$\vec{F}_\mu = -\mu F_N \operatorname{sgn}(\dot{\theta}) \hat{\theta}$$

$$\frac{\partial \vec{r}}{\partial n} = \hat{n}, \quad \frac{\partial \vec{r}}{\partial \theta} = R \hat{\theta}$$

$$Q_n = \vec{F} \cdot \hat{n} = -F_N - mg \cos \theta$$

$$Q_\theta = \vec{F} \cdot R \hat{\theta} = R(F_p - mg \sin \theta - cR\dot{\theta} - \mu F_N \operatorname{sgn}(\dot{\theta}))$$

radial:

$$-mR\dot{\theta}^2 = -F_N - mg \cos \theta$$

$$F_N = |mR\dot{\theta}^2 - mg \cos \theta|$$

angular:

$$mR^2\ddot{\theta} = R(F_p - mg \sin \theta - cR\dot{\theta} - \mu F_N \operatorname{sgn}(\dot{\theta}))$$

$$mR^2\ddot{\theta} = R(F_p - mg \sin \theta - cR\dot{\theta} - \mu(mR\dot{\theta}^2 - mg \cos \theta) \operatorname{sgn}(\dot{\theta}))$$

$$\ddot{\theta} = \frac{F_p}{mR} - \frac{g}{R} \sin \theta - \frac{c}{m} \dot{\theta} - \mu \left(\dot{\theta}^2 - \frac{g}{R} \cos \theta \right) \operatorname{sgn}(\dot{\theta})$$